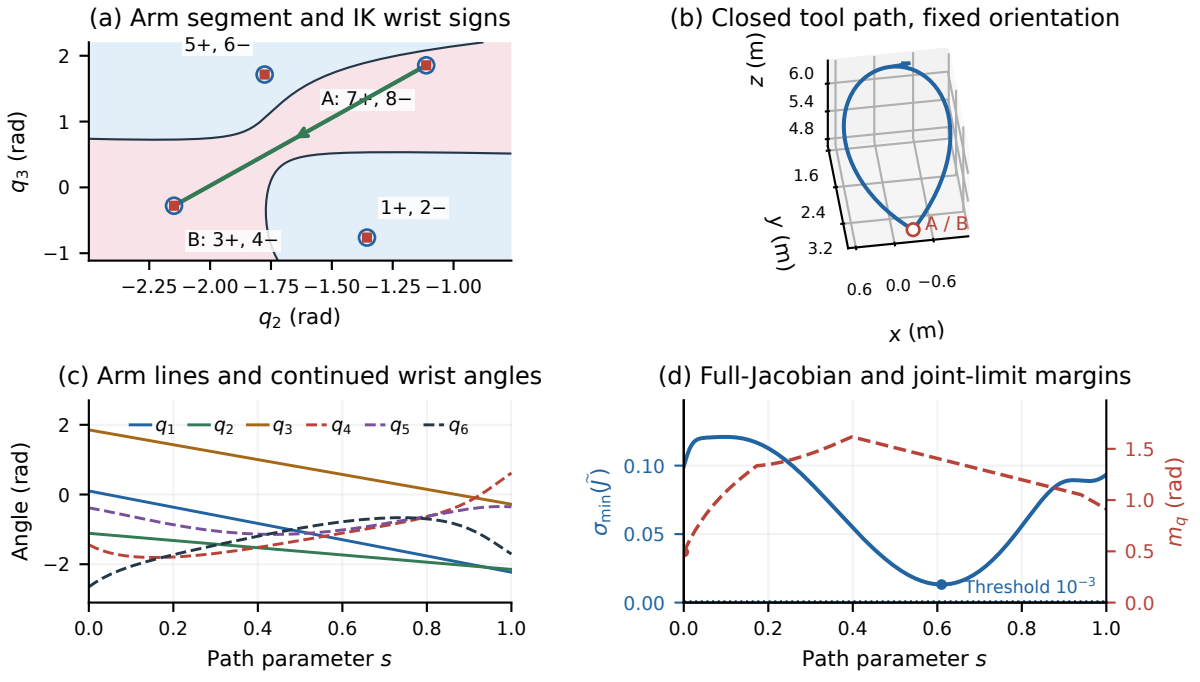

(a) Arm segment and IK wrist signs
5+, 6−
A: 7+, 8−
1+, 2−
B: 3+, 4−
q_3 (rad)
q_2 (rad)
(b) Closed tool path, fixed orientation
z (m)
y (m)
x (m)
A / B
(c) Arm lines and continued wrist angles
q_1 q_2 q_3 q_4 q_5 q_6
Angle (rad)
Path parameter s
(d) Full-Jacobian and joint-limit margins
$\sigma_{\min}(\tilde{J})$
m_q (rad)
Threshold 10^{-3}
Path parameter s